INCH-POUND

MIL-S-24711(SH)
AMENDMENT 2
17 September 1992
SUPERSEDING
AMENDMENT 1
21 December 1989

MILITARY SPECIFICATION

SWITCH, PROXIMITY, SOLID-STATE

This amendment forms a part of MIL-S-24711(SH), dated 27 October 1988, and is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense

PAGE 1

1.2 Delete and substitute:

"1.2 Classification. Solid-state proximity switches shall be classified as follows:

XX - Y Y - ZZZZ

XX - Type of input voltage
Y Y - Maximum sensing distance.
ZZZZ - Output states."

1.2.1, second sentence Delete "impact" and substitute "input".

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* 2.1.1, Under SPECIFICATIONS, MILITARY: Delete "MIL-E-917".

* 2.1 1, Under STANDARDS, MILITARY: Delete "MIL-STD-454".

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* 3.2.1.1, line 4: Delete "type V" and substitute "type I".

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* 3.2.1.5.1· Delete and substitute:

"3.2.1 5.1 Series connection All switch designations shall operate with two identical switches in series. The total switch voltage drop at the load shall not exceed 8.0 Vac rms at rated load current."

* 3.2 1.5 2· Delete and substitute:

"3.2.1.5.2 Parallel connection. All switch designations shall operate with two identical switches in parallel. The total switch leakage current through the load shall not exceed 4 0 mA rms."

* 3 2.1.6, line 6 Delete "temperature" and substitute "time".

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* Add as new paragraph

"3 2.2.10 Multiple switch operation Multiple switch operation shall be as specified (see 4 5 1) "

* Add as new paragraph

"3.2.2.10.1 Series connection All switch designations shall operate with two identical switches in series The total switch voltage drop at the load shall not exceed 3.0 Vdc at rated load current."

* Add as new paragraph

"3.2.2.10.2 Parallel connection All switch designations shall operate with two identical switches in parallel. The total switch leakage current through the load shall not exceed 600 microamperes.

* Add as new paragraph:

"3 2 3 1 Measurements Measurements shall be made using standard targets as specified in 3.2 3.4.1."

* Add as new paragraph:

"3 2 3 2 Head-on mode.

* Add as new paragraph

"3.2.3.2.1 Sensing distance. The sensing distances for the switches shall be as follows. (see 4 5.1)

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(a) For all 1.5-inch switches, the sensing distance in the head-on mode shall be 1.50 plus 0.50 minus 0 12 inches.

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(b) For all 0.5-inch switches, the sensing distance in the head-on-mode shall be 0.5 inches plus 0.05 inches minus 0.05 inches

The sensing distance of the switches shall operate as specified (see 4.5.1)."

* 3.2.3.2: Delete and substitute:

"3.2.3.2.2 Hysteresis. Hysteresis distances for the switches in the head on mode shall be as follows· (see 4.5.1)

(a) For all 1.5-inch switches, a minimum of 5 percent and a maximum of 40 percent of the nominal sensing distance.
(b) For all 0.5-inch switches, a minimum of 5 percent and a maximum of 20 percent of the nominal sensing distance."

* 3.2.3.3: Delete and substitute:

"3 2.3.3 Slide-by mode.

* 3.2.3.3.2: Delete and substitute:

"3.2 3.3.2 Actuation curve slope requirements The slope of the actuation curve shall be as steep as possible (see figures 2 and 3). Slope of the actuation curve for 1.5 inch switches shall not be less than 45 degrees between 0.5 and 1.0 inches above the sensing face in the slide-by mode while moving the target in planes parallel to the sensing face This slope shall be determined from measurements made at 0.50, 0.75 and 1 00 inches above the sensing face "

* Add as new paragraph:

"3.2.3.3.3 Hysteresis Hysteresis distances for the switches in the slide-by mode shall be as follows: (see 4.5.1)

(a) For all 1.5-inch switches, a minimum of 0.025 inches and a maximum of 1.12 inches in parallel planes up to 1.12 inches above the sensing face.
(b) For all 0.5-inch switches, a minimum of 0.01 inches and a maximum of 0.5 inches in parallel planes up to 0.38 inches above the sensing face "

* 3.2 3.4.1, line 2: Delete "as specified in MIL-S-22698" and substitute "in the fully annealed condition."

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* 3.2.7.2: Delete "0 to 100 percent humidity" and substitute "accordance with MIL-STD-202, method 103B, test condition B".

* Add as new 3.3.2 through 3.3.2.3.

"3.3.2 Prohibited materials.

3.3.2.1 Asbestos. Asbestos shall not be used.

3.3.2.2 Cadmium Cadmium shall not be used

3 3 2 3 Mercury Mercury in any form shall not be used in shipboard equipment, including materials and parts thereof Mercury shall not be used in manufacturing and test processes (including test equipment such as mercury indicators) applying to the basic equipment, but may be used in the manufacturing and test processes for materials and parts provided it is used in such a way that the materials and parts themselves cannot be contaminated No instruments containing mercury shall be used in the manufacture or testing of any equipment destined for installation on a nuclear powered ship "

* 3 4· Add to end of paragraph

"Tolerance for CE03 broadband emission test between 0.015 MHz and 0.05 MHz shall be 5 dBuA/MHz "

PAGE 12

4 5 2, line 3 Add "10" between "be" and "actuation".

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* 4 5 7, line 5, Add after "exposed": "and the switch housing connected to circuit ground."

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* 6.2 1 (b): Delete and substitute:

"(b) Designation of switches required (see 1.2.1)."

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Delete "FIGURE 1" and substitute "New FIGURE 1"

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* Delete "FIGURE 4" and substitute "New FIGURE 4"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:14Z
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* Delete "FIGURE 5" and substitute "New FIGURE 5".

6.8 Changes from previous amendment. The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment

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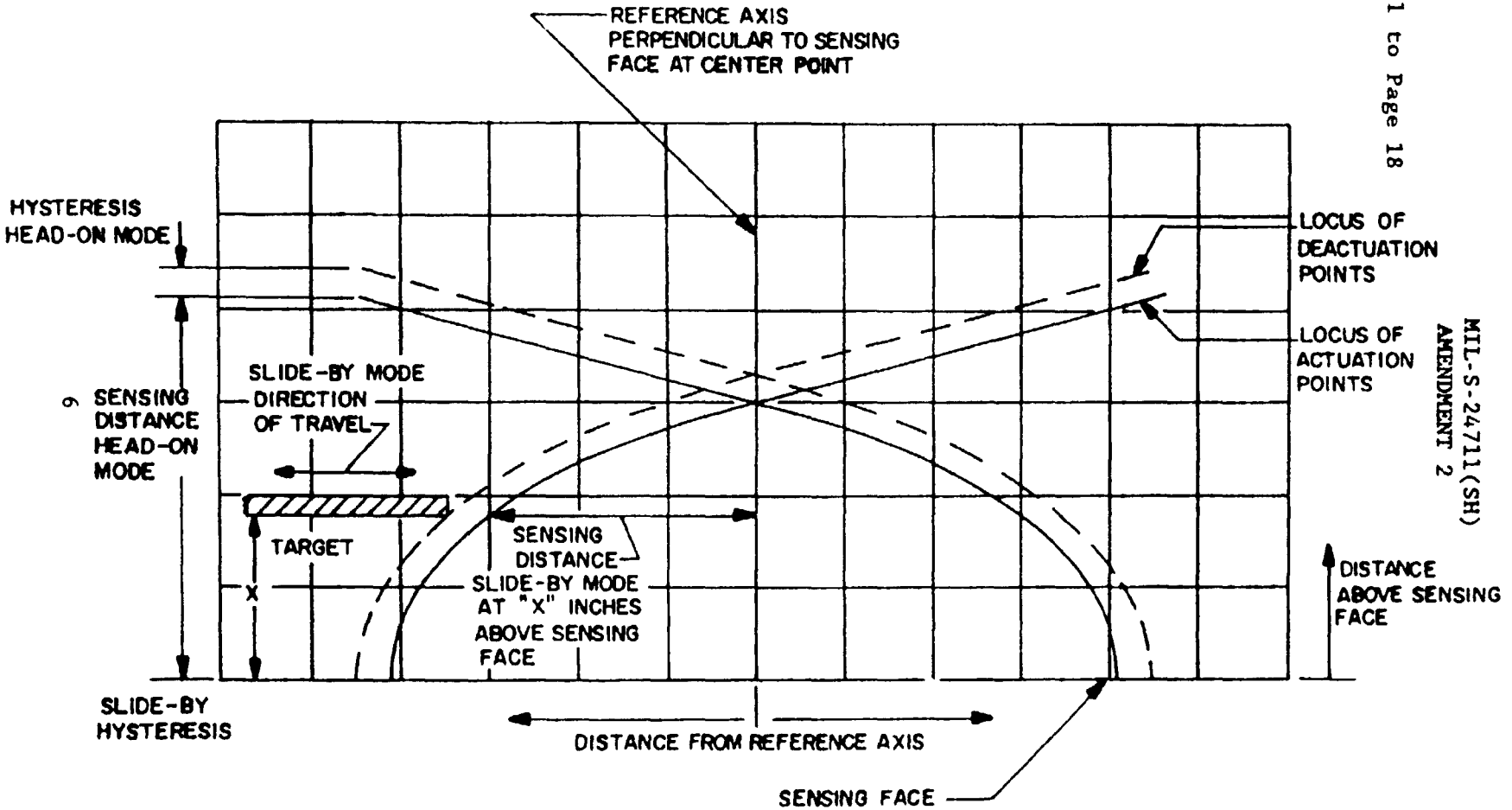


FIGURE 1 Typical actuation curve with hysteresis.

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New FIGURE 4 to Page 21

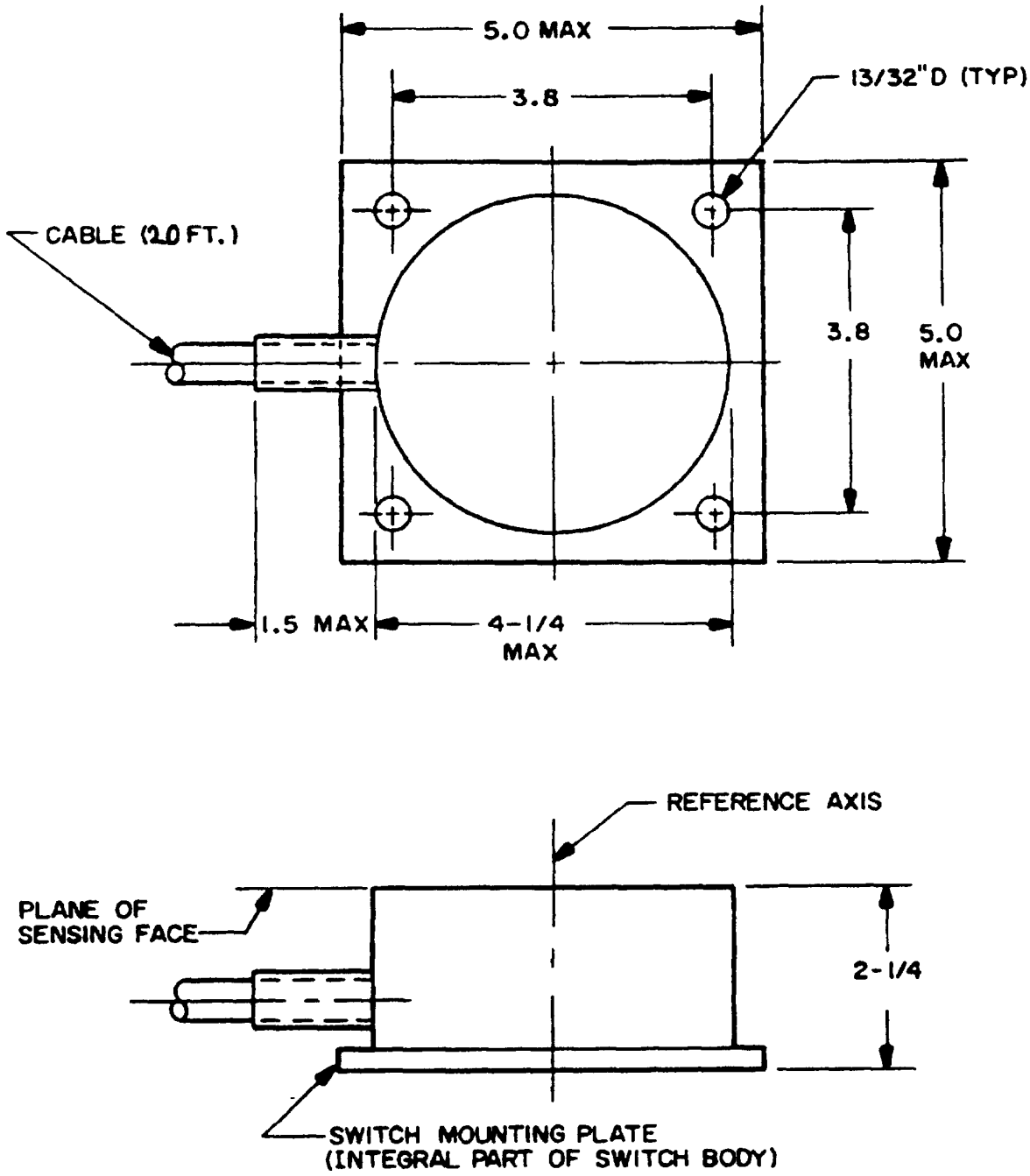
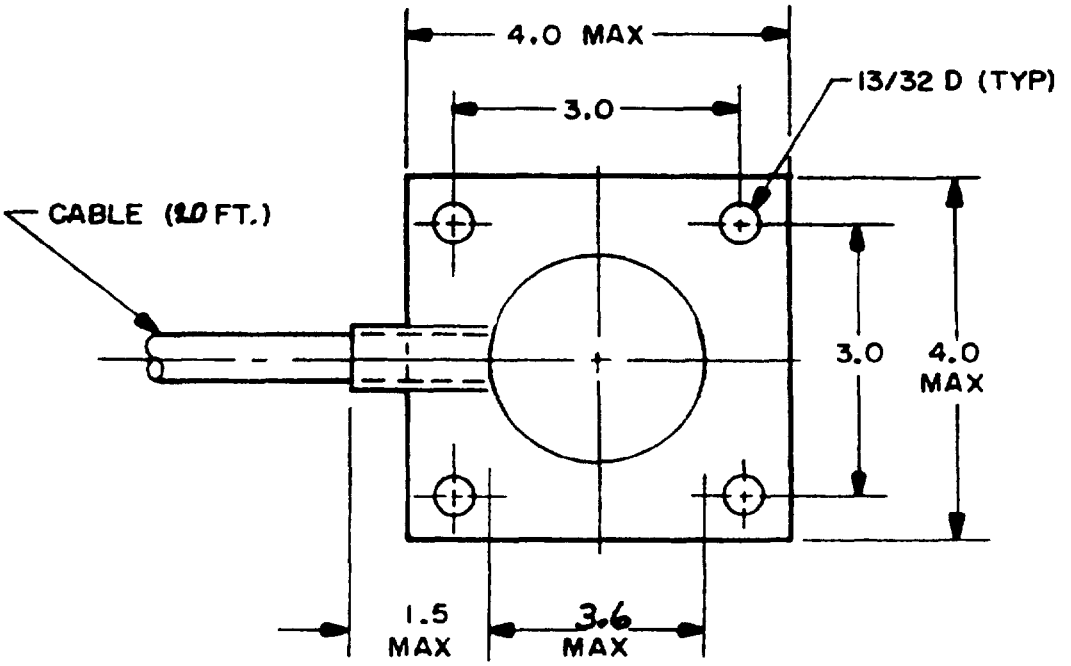
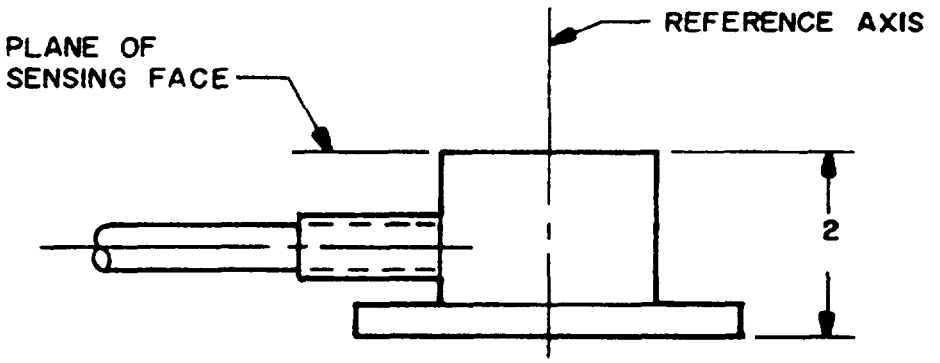


FIGURE 4. Outline dimensions for dc - 1.5-NONC, ac-1.5-NO, or ac-1.5-NC switch."

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New FIGURE 5 to Page 22





SWITCH MOUNTING PLATE
(INTEGRAL PART OF SWITCH BODY)

FIGURE 5 Outline dimensions for dc-0.5-NONC, ac-0.5NO, or ac-0.5-NC switch."

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